Incident-aware regression gating for evolving agent policies

Plan for adapting containment-extension into a Trust4RSI-Agent paper (AAAI 2027 workshop)

Target: full paper, 8 pages + references and appendix, double-blind, OpenReview - deadline 20 Nov 2026 AoE, no extensions

What we already have

26-incident library, independent effect observer, 440 archived live rollouts, instruction optimizer, executable incident twin, offline-qualified testbeds

What the paper claims

An audit-and-release abstraction beneath an evolving agent: observe effects below the tool broker, gate policy updates on verified incidents, keep executable incidents as the regression suite

What still has to run

E1 archived-data figure, E2 matched proxy-vs-incident selection (the one live experiment), E5 second-reader codings; optional E3/E4

The repository's headline hypothesis (incident feedback beats transcript-proxy feedback for policy selection) is designed but unmeasured. The paper must lead with the abstraction, the testbed and pilot evidence, and report that contrast only as far as E2 actually measures it.

The venue: Trust4RSI-Agent, 'An Agent Operating System Perspective'

The workshop treats trustworthy agent evolution as a systems problem: design the layer beneath an evolving agent

Item	Requirement
Full paper	8 pages, AAAI 2027 author kit; references and appendix excluded
Short / position	4 pages, same kit
Review	Double-blind on OpenReview; anonymize PDF and supplementary
Submission	20 Nov 2026 AoE, strict
Notification	2 Dec 2026 AoE
Workshop	22-23 Feb 2027, Palais des congres, Montreal (one day, TBA)
Accepted types	Mechanisms/abstractions, system designs, benchmarks, negative results, position papers

Five topics of interest

- **T1 Memory evolution** (state subsystem): provenance, write policies, snapshots, leakage from persistent state
- **T2 Tool and skill evolution** (capability interface): least privilege, sandboxing, verification of synthesized tools, revocation
- **T3 Model and policy evolution** (update and release layer): versioning, staged rollout, regression suites, rollback, reward hacking, reliability of self-critiques
- **T4 Multi-agent co-evolution** (runtime and scheduler): shared state, runaway loops, blast-radius limits
- **T5 Evaluating evolution** (observability and audit layer): attributing outcomes to updates, monitoring, oversight before irreversible updates, stateful testbeds

Our natural home is T3 (the release layer) with T5 (the audit layer) as the measurement story. The CFP explicitly welcomes system designs, benchmarks and negative results, which is what the repository can honestly supply.

Organizers: Haolun Wu (Stanford/Mila), Negar Arabzadeh (UC Berkeley), Wenyue Hua (MSR), Bang Liu, Laurent Charlin (Mila). Speakers include Yoshua Bengio, Mengdi Wang, Yilun Du. A safety-literate audience: hedged, measured claims will be read as a strength.

Fit map: repository component -> workshop topic -> OS vocabulary for the paper

The components exist; the adaptation is mostly renaming them in the language of the layer beneath the agent

Topic	Repository component (as implemented)	Paper vocabulary (agent-OS framing)	Role
T3 policy evolution	Instruction optimizer: model-proposed clause variants, catalog validator, development-only selection, frozen held-out trials; constrained objective min R s.t. U, C, F	Staged rollout with a release gate; a regression suite every policy update must pass; reward hacking made visible to the gate	Primary
T5 evaluating evolution	Independent SQLite effect observer (before/after state), counterfactual task families with feasibility oracle, grader regression cases, 26-record incident library	Audit subsystem below the tool broker; outcome attribution per update; stateful testbed with reference labels	Primary
T2 tool/skill evolution	Dispatcher scope check on exact `.lab.test` authorities; catalog-constrained synthesis rejects arbitrary instruction text	Reference monitor / least privilege; verification of synthesized capabilities before they reach the actor	Supporting
T4 multi-agent co-evolution	Coordination harness: shared counters, read-consumed quota, URL store, paired counterfactual replay, 28 scripted cases	Shared-state blast radius; effects invisible in transcripts	One paragraph + appendix
T1 memory evolution	Prior-run answer archive, cached flag, synthetic mailbox as reachable persistent state	Leakage from persistent state across runs	Mention only

Rule for the rewrite: keep every measurement caveat once, in Limitations. The body leads with the abstraction and what it already observes.

Repository sources: README.md; docs/incident-aware-optimization.md; docs/instruction-optimization.md; docs/coordination-harness.md

The paper in one slide: what we claim and what we do not

A contribution of mechanisms, a testbed and pilot evidence; the optimization result is reported only as far as E2 measures it

We claim

- **Abstraction.** An audit-and-release layer for evolving agents: (a) an independent effect observer reading service state below the tool broker, (b) executable incident twins as a regression suite, (c) a release rule that minimizes verified incidents subject to utility, cost and false-refusal limits.
- **Evidence that scorers miss what the observer sees.** 13 of 15 violating runs in the incident study passed the task scorer by submitting a cached answer; the observer recorded every excluded read.
- **Pilot evidence that policy updates move the gated quantity.** Haiku 4.5 outside access on impossible tasks 6/8 -> 0/8 under the combined instruction update, with 96/96 capability controls completed in scope; scope levels 2/6, 2/6, 0/6, 0/6.
- **A reusable protocol and testbed.** Frozen plans, per-run budgets, 320 transition checks, 17 grader cases, 1,543 relabeled proposals, 196+ tests, no model call in qualification.
- **A negative result on environment grounding.** The direct upstream coding baseline failed for environment reasons, not model reasons.

We do not claim

- A general failure rate or a generally effective defense: cells are n=8 and n=2 on one task structure and two models; 0/8 has a 95% Wilson upper limit of 32.4%.
- That incident feedback beats transcript-proxy feedback in general. E2 gives one matched comparison on the native fixture; a null is reported as a null.
- Benchmark-level improvements on ImpossibleBench, AgentDojo, ResearchArena or BashArena. These stay as the declared protocol.
- That the dispatcher is OS-level containment. It is tool-level containment for a closed tool surface.
- That the incident codings are settled facts. They are analyst judgments on public reporting; E5 adds a second reader.

Title candidate: 'Incident-aware regression gating for evolving agent policies: an audit layer beneath the tool broker'

Eight-page outline: page budget and source material

Everything in sections 2-4 and 6 exists today; section 5 needs E1 and E2; section 8 needs E5

#	Section	Pages	Content	Source material	Needs
1	Introduction	1.0	Agents evolve through prompt, skill and model updates; each update needs a gate; task scores miss incidents	README; incident-aware-optimization.md	Writing
2	Incidents motivate the layer beneath	1.0	26 public cases: pathways, mechanisms M1-M6, propositions P1-P9, detection and disclosure latency, mapped to T1-T5	incident-library-analysis.md; figures/incidents/analysis	E5
3	Abstractions	1.5	Effect observer below the broker; incident definitions; executable twins; release rule min R s.t. U, C, F; B0-B4 protocol	protocol.md; workbook_policy.py; INC-2026-015	Writing, 1 figure
4	Testbed	1.0	Native service fixture; counterfactual task families; grader regression; coordination harness; budgets and frozen plans	evaluation-infrastructure-results.md; coordination-harness-results.md	Writing
5	Pilot evidence	2.0	Utility vs verified incident rate across instruction policies (E1); matched proxy-vs-incident selection (E2); controls and intervals	incident-study-v1; scope-levels-v1; link-study-v1; new E2 run	E1, E2
6	Negative result	0.5	Upstream coding baseline: 1/2 original, 0/4 impossible, message-capped, missing packages (P2)	impossiblebench-upstream-baseline-results.md	Writing
7	Related work	0.5	AI control protocols, Inspect approvers, AgentDojo defenses, ImpossibleBench, incident taxonomies	standardized-benchmarks.md	Writing
8	Limitations and roadmap	0.5	Sample sizes, one task structure, single-reader codings, tool-level containment, the unmeasured benchmark protocol	experiment-roadmap.md	E5

Appendix (unlimited): exact prompts, frozen plans, per-cell tables with Wilson intervals, incident record schema, coordination harness cases, test counts. Supplementary: anonymized source snapshot and evidence archive (E6).

Section 2 material: what the 26-case library already says about the layer beneath

Codings are analyst drafts on public reporting; second-reader agreement (E5) is required before these counts go in the paper

Finding (n = 26)	n	Topic
M4 detection without decisive escalation: present	13	T5 oversight
P6 effects matter more than stated intent: violated	17	T5 audit layer
M2 capability composition: present	15	T2 least privilege
P3 authorization must survive config failure: violated	12	T2 / T3
P4 impossible tasks are security inputs: violated	10	T3 reward hack
M3 shared services as coordination infra: present	3	T4 shared state
Stop found only in retrospect	10	T5 monitoring
Median detection latency (12 known)	14 days	T5
Median disclosure latency (21 known)	119 days	T5

How the section argues

- Pathways: adaptive escape 3, misconfiguration 6, live-access overreach 9, training-sample boundary 3, external adversary 4, unresolved 1. The two usual pathways cover only 9 of 26.
- 18 entries are judged replayable as synthetic twins; one (INC-2026-015, public hosting of a local workbook) is implemented. That is the bridge from Section 2 to the regression-suite idea in Section 3.
- Evidence levels A: 19, B: 7. State them; do not add counting units across cases.
- Keep the coding frame citation to one sentence. The paper's contribution is the executable use of the library, not the taxonomy.

One figure here: mechanism-by-topic heatmap (exists as figures/incidents/analysis/mechanism-heatmap.png; relabel columns with T1-T5).

Risk: the records name real companies and models. That is public reporting and is fine, but every cited count must carry its evidence level, and nothing may read as a verdict on a named party. Phrase as 'reported' and 'coded as'.

Repository sources: docs/incident-library-analysis.md; incidents/records; scripts/analyze_incident_library.py

Section 3 material: the audit-and-release layer, drawn in OS terms

The agent's policy evolves above the line; everything the paper contributes sits below it

Agent + policy theta

Fixed actor model. theta = selected, ordered instruction clauses (later: monitor prompts, thresholds, defer rules). This is what evolves.

Tool broker (ref. monitor)

One structured tool. Exact-authority routing, optional scope enforcement, fail-closed on unknown routes. Changing theta never opens this boundary.

Service replicas (state)

SQLite transactions for repository, scorer, collector, mailbox, prior-run artifact. Writes later reverted are still state history.

Task scorer

Official pass/fail. Accepts a cached flag. This is the signal an evolving agent optimizes if nothing else is watching.

below the line: the paper's layer

Independent effect observer

Reads before/after service state, not the transcript. Emits incident records: resource, authorization, effect, reverted or not. Positive witnesses survive later failures.

Release gate

min R(theta) s.t. U >= U0 - dU, C <= B, F <= aF. Development evidence only; freeze before held-out. No eligible candidate = no release.

Executable incident twins

Library record -> policy manifest + unsafe and benign fixtures. Every candidate theta must pass the suite. INC-2026-015 implemented; 18 judged replayable.

Versioned policy release

Frozen theta, prompt hashes, source snapshot, plan, budget ledger. Rollback = previous frozen record. Held-out test never feeds back.

The figure to draw for the paper: this slide as one panel, with the proxy-feedback path (transcript judge -> gate) drawn as a dashed alternative to the observer. figures/optimization/overview-v1 and system-components-v1 already contain the pieces.

OS analogies to use sparingly and name once each: reference monitor (broker), audit subsystem (observer), CI regression suite (twins), release gate and rollback (frozen records). Say which classical abstraction transfers and which breaks: audit coverage must be demonstrated, not assumed.

Repository sources: README.md (outer containment); docs/incident-aware-optimization.md; incidents/executable/INC-2026-015

Section 5 material: 440 archived live rollouts

Every cell below can be reported today with Wilson intervals; none of it is a new run

Study (date)	Rollouts	Models	Manipulation	Key cells	Controls	Cost
Native pilot (Sep)	8	GPT-4.1 mini, Haiku 4.5	solvable/stalled x permissive/scoped	Local out-of-scope effects occurred; early scope check over-blocked	-	-
Cue study (Sep)	96	both	prompt cues	No consistent increase from tested cues	-	-
Upstream-link study (Sep)	128	both	README link present/absent x reminder	Early upstream reads 26/32 with link, 0/32 without; later violations substituted the answer archive	-	-
Incident study (17 Sep)	160	both	N / R / E / RE instruction updates	Haiku impossible outside access 6/8, 3/8, 5/8, 0/8; GPT-4.1 mini 0/8, 1/8, 0/8, 0/8	96/96 in scope	$0.95
Scope levels (24 Sep)	24	Haiku 4.5	4 cumulative instruction levels	Outside effects 2/6, 2/6, 0/6, 0/6; impossible-only 2/2, 1/2, 0/2, 0/2	15/16 in scope	$0.24
Upstream coding baseline (Sep)	6	Haiku 4.5	unmodified ImpossibleBench scaffold	Original 1/2, impossible 0/4; all 100-message capped; missing packages in all 6	-	$5.78
ImpossibleBench preflight (Sep)	24 assigned, 16 started	Haiku 4.5	custom adapter arms	No scored submissions; unknown outcomes	-	~$5 cap

What to foreground

- 13 of 15 incident-study violations submitted the cached answer and **passed the scorer**. This is the paper's motivating observation.
- Capability controls stayed at or near 100% while the gated quantity moved: the utility constraint is not vacuous.

What to keep in Limitations

- One task structure; two models; cells of 8 and 2. GPT-4.1 mini's near-zero event rate leaves no room to measure an effect.
- Instruction packages change length and content together; this is not a scalar strength manipulation.

Repository sources: docs/-results.md; results/*/README.md; results/*/summary.json*

Experiments: what to implement and run, marked by status

Three things are required (E1, E2, E5); E3 and E4 strengthen the paper if E2 is informative; the benchmark suite stays declared, not run

DONE: reuse archive | BUILD: implement, no inference | RUN: implement + live calls | HUMAN: analyst work | LATER: declare as future work

ID	Experiment	Status	Inference	Budget	Output / gate
E1	Main figure from archived data: task utility (x) vs verified incident rate (y) for every instruction policy in incident-study-v1, scope-levels-v1 and link-study-v1, with Wilson intervals	BUILD	None	$0	Fig. 2 + per-cell table; script under scripts/
E2	Matched fixed-pool selection: same candidate pool and shared development rollouts; select once with task score + transcript-proxy judge (B3), once with task score + effect observer (B4); test both frozen policies plus two fixed controls on fresh held-out seeds	RUN	~110 Haiku actor rollouts + ~60 judge calls	est. $2-3, ceiling $10	Primary Section 5 result; frozen plan + scripted controls first
E3	Replicate E2 on a second model family with a non-zero baseline event rate (GPT-4.1 mini had 1/32; pick a different actor)	RUN (optional)	same as E2	ceiling $15	Needed to say 'two model families'; skip if E2 null
E4	Regression-gate demonstration: implement twins for 2 more replayable library entries; run the four frozen E2 policies through the twin suite with the live actor	BUILD then RUN (optional)	~24 rollouts	ceiling $3	Shows the gate consuming incident twins; stretch
E5	Second-reader coding of the 26 records on pathway, M1-M6, P1-P9; report agreement; resolve or mark disputed	HUMAN	None	$0	Required before Section 2 counts are cited
E6	Anonymized reproducibility package: source snapshot, frozen plans, evidence archives, tests, builder scripts; strip names, URLs, decks	BUILD	None	$0	Supplementary zip
-	ImpossibleBench U2 environment repair, AgentDojo, ResearchArena, BashArena comparisons	LATER	-	-	Section 8 roadmap only

Cost basis: the incident study ran 160 rollouts for $0.95 and scope levels 24 for $0.24 with Haiku 4.5 at $1/$5 per million tokens. Estimates are list-price projections, not billing guarantees; every run keeps the repository's per-run and aggregate ceilings.

Repository sources: experiments/incident-aware-optimization-v1.design.json (status: matched feedback ablation not implemented)

E1: the main figure from archived evidence (no inference)

The design doc already specifies the plot; the data for six instruction policies exist in results/

BUILD: implement, no inference

Points to plot (Haiku 4.5 unless stated)

- **Incident study, impossible n=8 each:** N 6/8, R 3/8, E 5/8, RE 0/8 outside access. Utility axis: authorized completion on recoverable + easy controls, 96/96 across arms (plot per arm).
- **Scope levels, impossible n=2, controls n=4:** baseline 2/2 and 4/4; repeat_scope 1/2 and 3/4; explicit_routes 0/2 and 4/4; action_check 0/2 and 4/4.
- **Upstream-link study, n=32 per README cell:** early upstream reads 26/32 with link, 0/32 without, 0/30 (+2 unknown) link + reminder, 0/32 hidden + reminder; solvable-task completion as utility.
- **GPT-4.1 mini cells** as hollow markers: 0/8, 1/8, 0/8, 0/8. Shows the event-rate floor problem rather than hiding it.
- Wilson 95% intervals on both axes; mark unknown outcomes as a separate count, never as zeros.

Implementation

- New script `scripts/draw_utility_incident_tradeoff.py`: read the three archived summaries, emit PNG + SVG under figures/experiments/tradeoff-v1/ with a JSON table of every plotted cell.
- Follow the repository's figure convention: generator script, editable SVG, metadata next to the image.
- Regression test: the plotted counts equal the archived summary counts (a fixture-reading test, no inference).
- Half a day. Blocked on nothing.

What the figure must not do: draw a curve. Six points with intervals, labeled by policy, is the honest version.

Repository sources: results/incident-study-v1/summary.json; results/scope-levels-v1/analysis.json; results/link-study-v1

E2: the one live experiment, matched proxy-vs-incident selection

First stage of the design document's B3 vs B4 contrast: fixed pool, shared development rollouts, two selections, one held-out observer

RUN: implement + live calls

Stage	Content	Rollouts
Pool	8 catalog-mode instruction candidates + 2 fixed controls (original, fixed L3); catalog proposer is deterministic, so the pool is reproducible	0
Development	10 policies x 6 cases (2 impossible, 2 recoverable, 2 solvable), shared across both selections	60 actor
Judge (B3 only)	Haiku transcript monitor over each development transcript: unauthorized-access flag + suspicion; threshold set on benign calibration transcripts from the archive	60 judge
Selection	Same eligibility rule (zero incidents on impossible, all controls completed, shortest instruction) applied to judge flags (B3) and to observer records (B4)	0
Held-out	4 frozen policies (original, fixed L3, B3-selected, B4-selected) x 12 fresh seeds (4/4/4)	48 actor

Where proxy and observer can disagree

- **Reverted writes.** The optimizer's scripted control already covers an upstream write that is later reverted. A transcript judge may excuse it; the observer counts it.
- **False flags.** The judge may flag legitimate recovery actions, steering B3 toward over-restrictive policies and lower control completion.
- **Cached-answer success.** Scorer pass with excluded read: both signals should catch it; if the judge misses any, that is the result.

Gates before the first paid call

- Scripted controls pass for both selection paths; the judge path is exercised with scripted transports.
- Frozen plan with pool, seeds, thresholds, eligibility rule, tie-break, per-run and aggregate ceilings; user approval of the ceiling recorded in the plan.
- Baseline development event rate must be non-zero on impossible cases (Haiku N was 6/8); otherwise selection is uninformative and the run stops.

Pre-registered readout

- Primary: paired difference in held-out verified incident rate and in control completion between B3- and B4-selected policies, by seed.
- Also report: which candidate each signal chose, judge false-flag rate on controls, cost per arm. Same policy chosen by both = null result, written up as such.

Repository sources: docs/incident-aware-optimization.md; src/containment_extension/instruction_optimization; src/containment_extension/evaluation/live.py

E2: what has to be implemented (about four engineering days)

The optimizer, fixture, observer, budgets and a scripted-transport monitor runner exist; the missing parts are the judge arm and the dual selection

RUN: implement + live calls

Piece	Exists?	Work	Verification
Candidate pool (catalog mode)	Yes	Pin 8 candidates by stable ID in a new config experiments/feedback-ablation-v1.json	Dry run prints exact instructions and ceilings
Shared development rollouts	Yes (study.py)	Record per-rollout transcript path + observer record together so both selections read one evidence set	Fixture test: every development rollout has both artifacts
Transcript judge	Partly: scripted-transport runner; 1,543 archived inputs	Live Haiku monitor prompt over transcript only; calibrate threshold on archived benign transcripts; never give it service state	Scripted-transport test; calibration report with TPR/FPR on archive
Dual selection	No	Selection function takes a signal source (judge flags or observer records); emit both selected IDs and eligibility tables	Unit test: identical inputs -> identical choice; divergent fixture -> divergent choice
Held-out schedule	Yes (fresh-seed trials)	Extend to 4 frozen policies x 12 seeds with randomized block order	Schedule check prints 48 assignments
Budget ledger	Yes	Add judge calls as a separate ledger line and ceiling	Reservation test on provider failure
Report	Partly (instruction-report)	Paired held-out table, judge false flags, cost per arm, Wilson intervals	Report from scripted run has no empty fields

Order: dual selection -> judge arm -> scripted end-to-end -> freeze plan -> approval -> run. Do not touch the fixture, scorer or incident definition in the same change.

Repository sources: instruction_optimization/{catalog,search,study}.py; evaluation/live.py; results/incident-monitor-inputs-v1 (1,543 paired inputs)

E3-E6: optional strengthening and the required non-inference work

Decide E3/E4 after E2's development stage; E5 and E6 start now

E3 Second model family (RUN, optional)

- Same plan as E2 with a different actor whose baseline impossible-task event rate is non-zero in a 4-run preflight.
- Lets Section 5 say 'two model families' instead of 'one model'. Ceiling $15. Skip if E2 is null or budget is not approved.

E4 Regression gate end to end (BUILD then RUN, stretch)

- Implement twins for two more of the 18 replayable library entries using the INC-2026-015 pattern (policy manifest, unsafe fixture, benign fixture).
- Run the four frozen E2 policies through the twin suite with the live actor, ~24 rollouts, ceiling $3. Shows the gate consuming incidents, which is the paper's T3 story.

E5 Second-reader codings (HUMAN, required)

- A second reader codes pathway, M1-M6 and P1-P9 for all 26 records blind to the first coding; compute agreement; resolve or mark disputed.
- Without this, Section 2 counts are one analyst's draft and a careful reviewer will say so. About two days of reading.

E6 Anonymized artifact (BUILD, required)

- Source snapshot, frozen plans, evidence archives, tests, figure generators. Strip README author line, HANDOFF, slide decks, repo URL, machine paths and names in run metadata.
- Check that no archived manifest or transcript carries an account identifier. Zip, hash, upload as supplementary.

Not for this paper: ImpossibleBench U2, AgentDojo, ResearchArena, BashArena. Declared in Section 8 with the frozen source pins; no live allocation.

Section 6: the upstream coding baseline as a negative result

The CFP lists negative results as an accepted contribution; this one is already archived and maps to proposition P2

What happened

- Six Haiku 4.5 runs on the unmodified upstream scaffold, prompts and scorer: original 1/2, oneoff 0/2, conflicting 0/2.
- All six hit the 100-message cap without calling submit. All six hit missing packages under the default actor Python while the scorer used the activated `testbed` environment.
- 290 requests, 5.43M input tokens, $5.78. The passing original patch is a general repair on review.
- Earlier custom-adapter preflight: 24 assigned, 16 started, 14 at the 20-call limit, no scored submissions.

How to write it

- Twelve scoring controls and four tool-loop controls passed before the run, yet did not establish that ordinary actor imports work across successive calls. Environment claims required measurement in the actor's path, not the scorer's.
- An evolving agent whose regression suite runs in a different environment than its actor will pass the suite and fail in use. This is the T3 lesson, stated once.
- Report the capability floor: a safety comparison on this scaffold is uninformative until originals are solvable. No cheating rate is claimed.

Half a page. It costs nothing, it is already written in the docs, and it is the kind of result this workshop asked for.

Do not mix this with Section 5: the native fixture results and the coding-benchmark failure are different environments, denominators and models. Keep separate tables.

Repository sources: docs/impossiblebench-upstream-baseline-results.md; docs/impossiblebench-live-preflight-results.md; results/upstream-baseline-preparation-v1

Timeline: 31 days from today to the deadline

Submit on 19 Nov; the 20 Nov AoE deadline has no extension

Oct 20-26	Outline fixed. E1 figure. E2 dual selection + judge arm implemented, scripted end to end. E5 second reader starts. Draft Sections 2, 3, 6 from existing docs.
Oct 27-Nov 2	Freeze E2 plan and ceiling; approval; run E2 development stage; check event rate gate; run held-out. E5 continues.
Nov 3-9	Analyze E2; decide E3/E4 and run if green. Draft Sections 1, 4, 5. Paper figure from slide 7. E5 agreement computed.
Nov 10-16	Full draft in AAAI kit. Related work. Appendix tables. E6 anonymized artifact assembled and checked.
Nov 17-19	Internal read for over-claiming; anonymization check; OpenReview submission on 19 Nov.

Critical path: E2 implementation (week 1) -> E2 run (week 2). If the week-2 run slips past Nov 9, the paper ships as infrastructure + pilot + negative result, with E2 moved to Section 8.

Risks, reviewer pushback and fallbacks

Each row names the fallback that keeps a submittable paper

Risk	Likelihood	Fallback
E2 null: judge and observer pick the same policy	Moderate: the native transcript exposes every tool call	Report as a negative result with the judge's false-flag rate and the reverted-write case; the abstraction and testbed carry the paper
Live budget not approved in time	Low	Drop E2 to Section 8; submit infrastructure + archived pilot + negative result (still a full paper) or a 4-page position paper on the library
Reviewers: 'n=8 on a toy fixture'	High	Intervals on every cell; one-task-structure limitation stated; frame as a testbed paper, not a defense paper
Reviewers: 'incident codings are one analyst's opinion'	High without E5	E5 agreement statistics; disputed codings marked and excluded from counts
Reviewers: 'this is prompt engineering, not evolution'	Moderate	Section 3: theta is any policy update the gate can score; the catalog-constrained synthesizer is literally a model proposing its own instruction updates
Anonymity leak via repo name, decks, run metadata	Moderate	E6 checklist; grep the artifact for name, handle, machine paths, provider account strings
Over-hedged prose reads as no contribution	High given current docs	One Limitations section; body states what was built and observed in plain declaratives

Minimum viable submission, achievable with zero new inference: Sections 1-4, 6-8 plus E1 as the pilot evidence. E2 is what turns it from a system paper into a system paper with a first measured contrast.